PRITAM KUMAR

Embedded Systems Engineer | Firmware Developer | IoT & RTOS Specialist

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PROFESSIONAL SUMMARY

Embedded Systems Engineer with hands on experience across multiple microcontroller families (STM32, ESP32, AT89S51, Nordic nRF52840), with experience developing BLE based firmware and drivers using Zephyr RTOS. Strong foundation in bare metal (bare metal) embedded C, interrupt handling, UART/I2C/SPI/GPIO protocols, FreeRTOS, and PCB design. Built a industrial automation system (custom Ladder Logic VM on STM32) and multiple IoT sensor systems, with focus on BLE FOTA architecture, secure bootloader design, and hardware firmware integration.

CORE COMPETENCIES

- **Microcontrollers & SoCs:** Nordic nRF52840 | STM32 | ESP32 | AT89S51 | NodeMCU
- **RTOS & Firmware:** Zephyr RTOS | FreeRTOS | Bare metal Embedded C | BLE & FOTA (MCUboot) | Driver Development | Bootloader Design
- **Protocols & Interfaces:** UART | I2C | SPI | GPIO | BLE | MQTT | SMP
- **IDEs & Tools:** Keil µVision | Arduino IDE | STM32CubeIDE | nRF Connect SDK | VS Code | CMake/West
- **Design & Simulation:** KiCad | Proteus | Multisim | Xilinx Vivado/ISE | MATLAB
- **Version Control:** Git | GitHub
- **Core Skills:** Hardware Firmware Integration | Interrupt & DMA Programming | PCB Design & Schematic | Debugging & Verification | Technical Documentation

WORK EXPERIENCE

Junior Research Fellow (JRF) Sep 2026 – Present

Indian Institute of Technology (IIT) Delhi | New Delhi, India

- Developing embedded firmware and hardware for a sponsored biomedical device research project involving real time sensor acquisition and actuator control.
- Advancing a working prototype from breadboard to custom PCB, with emphasis on reliability, power efficiency, and hardware firmware verification.

Graduate Engineer Trainee — R&D / New Product Development Jun 2026 – Sep 2026

Morphedo Technologies | Noida, India

- Develop Nordic nRF52840 bare metal and Zephyr based firmware and peripheral drivers within a defense oriented R&D environment.
- Apply Zephyr build system concepts — devicetree, Kconfig, CMake, and west workspace for new product bring up and active development cycles.
- Building hands-on depth in BLE based FOTA architecture using MCUboot (swap with revert mode) for secure, fail safe over the air updates.

Research Intern — Embedded Systems Jun 2025 – Jul 2025

National Institute of Technology (NIT) Patna | Patna, India

- Architected a multi directional ultrasonic echo locator on NodeMCU using three HCSR04 sensors multiplexed via a 74HC139 demultiplexer, with independent GPIO TRIG/ECHO lines and indicator LEDs per sensor.
- Integrated DHT11 temperature compensation to improve ranging accuracy across varying ambient conditions.
- Engineered round robin sensor switching(3 second intervals) ensuring only one sensor reports a valid reading at a time, preventing cross talk interference.
- Explored acoustic impedance based material classification using reflected ultrasonic signal characteristics as the core research objective.

KEY PROJECTS

STM32 Mini Industrial PLC *STM32 | Embedded C | I2C | SPI | GPIO | KiCad*

- Developed a compact industrial PLC on STM32 featuring a custom Ladder Logic interpreter and bytecode VM, enabling programmable relay logic control and hardware firmware integration testing.
- Engineered optocoupler isolated digital I/O with interrupt driven GPIO and relay outputs, along with an SH1106 OLED and AT24C64 EEPROM over I2C for program storage.
- Created a companion ESP32 web dashboard for remote monitoring and control of PLC I/O states over WiFi; produced complete technical documentation.

Flipper EDU — Embedded Security Toolkit *ESP32 | Embedded C | Arduino IDE | BadUSB*

- Developed a Flipper Zero inspired security toolkit on ESP32 with two independently verified attack modules on physical hardware.
- Implemented Wi Fi deauthentication via 802.11 deauth frames and a BadUSB module enumerating as a USB HID keyboard for autonomous keystroke injection, demonstrating wireless and HID vulnerabilities in a controlled lab environment.

EDUCATION

B.Tech — Electronics & Communication Engineering *2022 – 2026*

Central University of Rajasthan (CURAJ), Ajmer, Rajasthan *CGPA: 7.91 / 10*

Coursework: Embedded Systems, Microcontrollers, UART/SPI/I2C, Signal Processing, VLSI Design, System Software